Testleaf
Always Ahead

Classification Algorithm

A **classification algorithm** in supervised learning **predicts discrete categories or labels** based on input data. Rather than forecasting a continuous value, the model assigns each input to one of a fixed set of classes — making it one of the most widely used techniques in machine learning.

Core Concept

Classification maps input features to a **distinct category**. The algorithm learns decision boundaries from labeled training data, then applies them to new, unseen examples.

Classic Example: Email Filtering

One of the most intuitive applications is classifying emails as **spam** or **not spam**. The model analyzes features like subject line keywords, sender reputation, and link patterns to make a binary prediction.

Discrete Output

Unlike regression, classification always produces a label — a category from a predefined set such as "spam/not spam" or "cat/dog/bird."

Supervised Learning

The algorithm trains on labeled examples, learning the patterns that distinguish one class from another before making predictions on new data.

Real-World Applications

From fraud detection and medical diagnosis to sentiment analysis and image recognition — classification powers decisions across every industry.